
REWARDING THE GRAPH BEHIND THE CHAIN: TOPOLOGY-AWARE PROCESS SUPERVISION FOR RL AND REASONING DISTILLATION

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ABSTRACT

Long chain-of-thought reasoning is not purely linear: intermediate results are reused across non-adjacent steps, while unsupported conclusions may refer backward or form circular support. Outcome rewards and step-local process rewards do not expose these failures. We propose TopoPRM, a topology-aware implicit process reward model that first recovers an order-agnostic directed support graph from a free-form trace and then projects it onto a dependency DAG. Crucially, the backward and cyclic edges removed by this projection are retained as direction-consistency and acyclicity signals. We combine outcome, format, topology, continuity, and length through a topology-conditioned Choquet reward whose interaction terms explicitly value agreement between structural and non-structural sources. The same projected graph guides distillation toward a compact dependency-preserving trace, without requiring graph-format generation from the policy. We evaluate whether TopoPRM advances the fixed-scale accuracy–length Pareto frontier over outcome-only GRPO and additive or multiplicative aggregation, together with controlled edge-recovery, topology-perturbation, and reward-hacking tests.

1 INTRODUCTION

Reinforcement learning with verifiable rewards (RLVR) has substantially advanced long chain-of-thought (CoT) reasoning in large language models and now serves as a core mechanism for training mathematical and symbolic reasoning ability (Wei et al., 2022; Kojima et al., 2022; DeepSeek-AI, 2025; Yeo et al., 2025; Shao et al., 2024; Ke et al., 2026). From sparse final-answer rewards (Cobbe et al., 2021; DeepSeek-AI, 2025) to widely used process reward models (PRMs) that attach supervision to intermediate positions or prefixes (Lightman et al., 2023; Wang et al., 2024; Uesato et al., 2022), the dominant training interface still represents a trace as a *linear sequence of steps*. In practice, long reasoning is rarely linear: in mathematical derivations, a later conclusion often depends on several non-adjacent premises, an intermediate equation is reused by multiple downstream derivations, and subgoals branch before merging into the final answer (Besta et al., 2024; Zhong et al., 2026). Reconciling this structural mismatch has emerged as a core challenge for advancing reasoning supervision beyond linear credit assignment.

Recent reasoning models have made the macroscopic structure of inference clearer and more reliable, yet they remain confined to the CoT-as-result view. DeepSeek-R1 (DeepSeek-AI, 2025), DAPO (Yu et al., 2025), and Tulu 3 (Lambert et al., 2024) demonstrated that outcome-only rewards scale, but they could not distinguish a well-supported derivation from a trace that reached the same answer through unsupported intermediate claims, and they remained vulnerable to reward hacking (Gao et al., 2023; Skalse et al., 2022). On the PRM side, denser signals have been obtained through Monte Carlo rollouts (Wang et al., 2024; Luo et al., 2024), trained verifiers (Zhang et al., 2025; Khalifa et al., 2025; Zhao et al., 2026), and implicit outcome-derived scoring (Yuan et al., 2025; She et al., 2025). These methods can condition on a full prefix, but they do not explicitly recover the **multi-parent support relation** in which one intermediate result supports, or is supported by, several non-adjacent steps (Wang et al., 2025). Likewise, token-level distillation (Agarwal et al., 2024) does not require a compact target to preserve those cross-step dependencies.

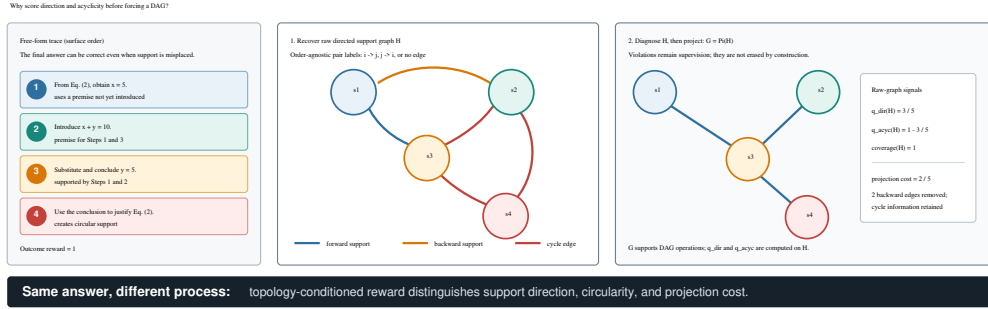


Figure 1: Why raw directed support must be scored before acyclic projection. A correct answer can still contain backward dependencies or circular justification. We retain these violations in $\mathcal{H}_y$, compute direction and acyclicity there, and only then form the DAG $\mathcal{G}_y = \Pi(\mathcal{H}_y)$.

We propose TOPOPRM, which supervises reasoning through its *implicit topological dependency structure* rather than assuming that surface order is already valid (Figure 1). A frozen, order-agnostic edge encoder first recovers a raw directed support graph $\mathcal{H}_y$ from an unchanged free-form trace. Projecting $\mathcal{H}_y$ onto a text-order DAG $\mathcal{G}_y$ makes the discarded structure informative: backward-edge mass measures direction inconsistency, while cycle-edge mass measures circular support. These diagnostics then enter a topology-conditioned, non-additive aggregation of outcome, format, topology, continuity, and length. The interaction terms reward agreement between topology and the other sources instead of reducing topology to a small additive bonus. Finally, the same projected graph guides distillation toward a compact dependency-preserving trace. The policy never has to decode graph tags or a special DAG format. Our contributions are threefold:

1. **Raw support graph before acyclic projection.** We recover directed dependencies without imposing textual order, and turn the backward and cyclic structure removed by projection into explicit process supervision.
2. **Topology-conditioned multi-source optimization.** We introduce a monotone 2-additive Choquet reward whose topology–outcome, topology–continuity, and topology–length interactions cannot be represented by an ordinary weighted sum, and reuse the projected graph for Topology-Guided Distillation.
3. **Accuracy–efficiency evaluation.** We compare against outcome-only and explicit length-shaped GRPO, projected-graph controls, additive and multiplicative aggregation, and distillation baselines under a fixed-scale, fixed-budget protocol, reporting both accuracy and the accuracy–length Pareto frontier.

2 RELATED WORK

Reinforcement Learning with Verifiable Rewards. RLVR replaces preference models with programmatic correctness checks on tasks whose answers are objectively verifiable (Shao et al., 2024; DeepSeek-AI, 2025; Lambert et al., 2024; Zeng et al., 2023). Recent work refines GRPO-style optimization through dynamic sampling and clipping (Yu et al., 2025), extended training schedules (Liu et al., 2025), finer credit assignment (Kazemnejad et al., 2025), asymmetric handling of negative samples (Zhu et al., 2025), entropy regularization (Cui et al., 2025), and simplified baselines (Ahmadian et al., 2024; Hu, 2025). Outcome-only signals nevertheless remain structurally blind: traces with very different dependency support can be reward-equivalent (Chen et al., 2026), and reward hacking persists (Gao et al., 2023; Skalse et al., 2022). *TopoPRM injects frozen topology and continuity diagnostics from latent reasoning graphs, targeting structural credit assignment directly rather than through a correctness surrogate.*

Concise-reasoning objectives introduce a separate confound: penalizing length inside a group-relative reward can assign negative advantage to a correct but long rollout. DRPO addresses this by decoupling the length-driven target distribution from correctness discrimination (Li et al., 2026). Our claim is not that topology replaces this optimization issue; we therefore compare against a fixed

outcome–format–length reward under the same completion budget and treat accuracy and length as co-primary outcomes.

Process Reward Models for Reasoning. PRMs score intermediate steps to address the sparsity of outcome rewards, originally relying on dense human annotation (Uesato et al., 2022; Lightman et al., 2023). Subsequent work reduces labeling cost via Monte Carlo or tree-search rollouts (Wang et al., 2024; Luo et al., 2024; Zhang et al., 2024; Lu et al., 2024; Zhang et al., 2025; Guan et al., 2025), generative PRMs make verifier reasoning explicit (Khalifa et al., 2025; Zhao et al., 2026; She et al., 2025), and process advantage verifiers tie step values to rollout-induced returns (Setlur et al., 2025; Lai et al., 2024). Implicit PRMs derive step-level signals from outcome feedback alone (Yuan et al., 2025; Snell et al., 2024), yet recent benchmarks expose persistent weaknesses on fine-grained error categories and inter-step consistency (Song et al., 2025; Zheng et al., 2025). Although such scorers may read the full prefix, their output is attached to a position in one sequence rather than to an explicit many-to-many support relation among steps. *TopoPRM trains no scalar neural process verifier and claims no proof-level guarantee; it instead derives frozen global-structure supervision from a recovered dependency graph together with local continuity.*

Structured and Graph-Based Reasoning. PARC links a completed chain to preceding premises for error identification (Mukherjee et al., 2025), while GRiD generates knowledge/reasoning nodes and verifies each node against declared parents (Wen et al., 2025). Other methods require DAG-formatted reasoning: DAG-Math evaluates graph-level logical closeness (Zhang et al., 2026), Structured Reasoning rewards within-graph flow and cross-sample overlap (Dong et al., 2026), Structural CoT trains executable computation graphs (Anonymous, 2026b), and the Graph Reasoning Paradigm emits labeled symbolic graphs (Liu et al., 2026). RewardFlow instead propagates terminal reward across an agent state graph (Feng et al., 2026); Graph-GRPO propagates credit over a predefined e-commerce relevance graph (Che et al., 2026). TopoPRM leaves policy traces in free text and recovers an unconstrained directed support graph before projection. Preceding-link and valid-DAG representations rule out backward and cyclic support by construction; we retain these residuals as direction and circular-justification signals, then couple them non-additively to outcome, continuity, and length.

Post-hoc graph probes support non-sequential reasoning (Zhong et al., 2026; Matsutani et al., 2026); CoT2Graph uses recovered graphs for failure analysis, outcome prediction, and Best-of- N selection (Anonymous, 2026a). Hidden-state cycles can also reflect useful recurrent visits (Minegishi et al., 2025). Our cycles instead denote circular semantic support in the committed derivation: we do not penalize exploration or revision, only a final support relation in which a claim depends on itself. The novelty is therefore not “reasoning as a graph,” but raw direction/cycle diagnostics as an online topology-conditioned multi-source reward without graph-formatted generation.

Distillation and Reasoning Compression. Knowledge distillation transfers teacher capability (Hinton et al., 2015); student-generated variants reduce train–test mismatch (Gu et al., 2024; Agarwal et al., 2024; Ko et al., 2024), including ExOPD, BOND, KDRL, and black-box GAD (Yang et al., 2026; Sessa et al., 2024; Xu et al., 2025; Ye et al., 2025). Long-CoT compression uses pruning, length penalties, or budget control (Xia et al., 2025; Luo et al., 2025; Arora & Zanette, 2026; Aggarwal & Welleck, 2025; Muennighoff et al., 2025); graph-based pruning removes weak branches from projected DAGs (Yuan et al., 2026), while Reasoning Scaffolding distills semantic operations (Wen et al., 2026). TopoPRM instead uses raw direction/cycle violations to elicit and filter teacher revisions, preserving dependencies that token budgets or post-projection pruning cannot identify. *Topology-Guided Distillation is evaluated by the accuracy–length frontier and independently audited topology preservation.*

3 METHOD

TopoPRM (Figure 2) is a topology-aware post-training framework whose stages share a graph encoder $\mathcal{E}_\phi$ and an acyclic projection Π . Stage I produces parsable reasoning traces. Stage II applies GRPO to a topology-conditioned multi-source reward. Stage III performs Topology-Guided Distillation, in which the Stage-II teacher revises student rollouts using defects localized on the projected graph.

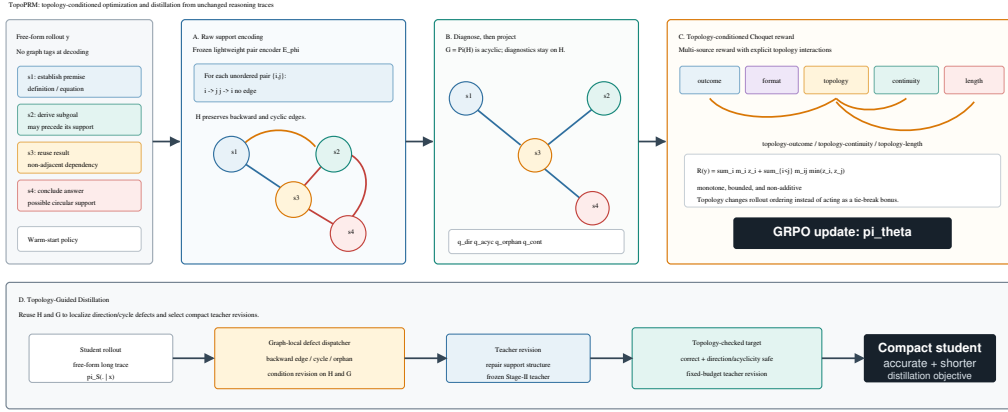


Figure 2: Overview of TopoPRM and Topology-Guided Distillation. An order-agnostic encoder recovers a raw directed support graph; its acyclic projection provides the training DAG while discarded backward and cyclic edges remain reward signals. A topology-conditioned Choquet reward aggregates multiple sources, and the projected graph guides dependency-preserving distillation.

3.1 PROBLEM SETUP

Let $x = (q, c)$ pair a mathematical problem q with optional context c , and let π_θ generate a trace $y = (s_1, \dots, s_T)$. The frozen encoder produces a raw directed support graph $\mathcal{H}_y = \mathcal{E}_\phi(y) = (\mathcal{V}, \mathcal{A}_H)$, where $i \rightarrow j$ means that step s_i supplies a result, claim, sub-goal, or assumption used by s_j . Unlike a DAG constructed only from pairs $i < j$, $\mathcal{H}_y$ may contain backward edges and cycles. We retain these violations and separately form $\mathcal{G}_y = \Pi(\mathcal{H}_y)$ by keeping support edges consistent with the trace order. Thus $\mathcal{G}_y$ supports DAG operations, while the projection cost of $\mathcal{H}_y$ exposes structural failures that would otherwise be removed by construction.

Scope of the acyclicity assumption. Acyclicity is normative only for the *semantic support relation in the committed trace*. It does not forbid exploratory computation, checking an earlier result, or revising a failed branch. Such behavior may revisit a latent state or repeat a concept while still yielding an acyclic final justification. A cycle in $\mathcal{H}_y$ has the narrower meaning that a written claim ultimately supports itself; q_{acyc} targets this circular justification rather than recurrent reasoning in general.

3.2 TOPOLOGY-AWARE DATA PROCESSING

Raw-Graph Extraction and Projection. Given a rollout y , $\mathcal{E}_\phi$ proceeds in three stages. **(i) Segmentation and parsing.** It splits y into steps and sub-questions, then extracts normalized expressions, claim keys, variables, and step types. **(ii) Order-agnostic support encoding.** Auditable rules propose high-precision symbolic candidates, while a lightweight pair encoder predicts one of $\{i \rightarrow j, j \rightarrow i, \emptyset\}$ for each unordered intra-component pair. The encoder is trained offline from semantic-teacher edges and frozen during policy optimization; a separately annotated set calibrates edge precision and direction. Retained encoder edges carry their calibrated confidence, while rule edges use unit weight. Semantic predictions override an order-based rule when both judge the same pair, preventing artificial two-cycles caused by conflicting extractors. **(iii) Acyclic projection.** We preserve all predicted edges in $\mathcal{H}_y$, set $\mathcal{G}_y = \Pi(\mathcal{H}_y)$ by removing $i \rightarrow j$ with $i \geq j$, and add an adaptive weak sequential edge only when a node remains unsupported. Full rules and abstention thresholds are in Appendix C.

Topology and Continuity Diagnostics. Let $\mathcal{A}_H^{\text{dep}}$ denote the dependency edges of $\mathcal{H}_y$, let $w_{ij} \geq 0$ be the confidence weight of edge (i, j) , and set $W_H = \sum_{(i,j) \in \mathcal{A}_H^{\text{dep}}} w_{ij}$. Let $\mathcal{A}_{\text{cyc}}$ contain the dependency edges whose endpoints lie in the same cyclic strongly connected component (size greater than one, or a self-loop), and let c_H be the fraction of nodes incident to at least one dependency edge. We first compute d_H , the forward-edge mass, and u_H , the cycle-edge mass, then shrink both

diagnostics toward 0.5 when edge coverage is low:

$$q_{\text{dir}}(\mathcal{H}_y) = c_H d_H + (1 - c_H)/2, \quad d_H = \frac{\sum_{(i,j) \in \mathcal{A}_H^{\text{dep}}} w_{ij} \mathbb{1}[i < j]}{W_H}, \quad (1)$$

$$q_{\text{acyc}}(\mathcal{H}_y) = c_H(1 - u_H) + (1 - c_H)/2, \quad u_H = \frac{\sum_{(i,j) \in \mathcal{A}_{\text{cyc}}} w_{ij}}{W_H}. \quad (2)$$

When $W_H = 0$, we use $d_H = 1$ and $u_H = 0$; because $c_H = 0$ for a multi-step empty graph, both coverage-adjusted scores are nevertheless 0.5, not perfect. (A singleton trace has no possible cross-step violation and uses $c_H = 1$.) The topology score q_{topo} is a normalized combination of q_{dir} , q_{acyc} , non-orphan support on $\mathcal{G}_y$, and optional reference-edge agreement. The continuity score q_{cont} measures whether each step is traceable to the question or a supported prefix. Direction and acyclicity are therefore properties of the raw graph, not tautological checks on an already projected DAG. They are related but not redundant: every directed cycle contains a backward edge under the textual order, whereas a backward edge need not close a cycle, and raw graphs with identical backward-edge mass can have different cycle-edge mass. Thus q_{dir} detects local order inconsistency while q_{acyc} detects global circular support.

Topology-Conditioned Multi-Source Aggregation. Let $\mathbf{z} = (r_{\text{out}}, r_{\text{fmt}}, q_{\text{topo}}, q_{\text{cont}}, r_{\text{len}}) \in [0, 1]^5$. A weighted sum assumes that each source has the same marginal value regardless of the others. Instead, we use a monotone 2-additive Choquet model with Möbius coefficients

$$r_{\text{total}}(x, y) = \sum_i m_i z_i + \sum_{(i,j) \in \mathcal{I}_{\text{topo}}} m_{ij} \min(z_i, z_j), \quad \mathcal{I}_{\text{topo}} = \{(o, t), (t, c), (t, l)\}. \quad (3)$$

All $m_i, m_{ij} \geq 0$ and sum to one, so the reward is bounded and monotone. The three interaction terms assign additional value only when topology agrees with outcome, continuity, or compactness. This makes topology a constituent of cross-source utility rather than a generic bonus applied after correctness. We ablate the capacity against equal-weight, tuned additive, and multiplicative alternatives.

3.3 TOPOLOGY-GUIDED POST-TRAINING

Stage I: Supervised Warm-Start. We perform Supervised Fine-Tuning (SFT) on structured mathematical reasoning data to teach the policy to produce stepwise solutions parsable by $\mathcal{E}_\phi$. This reduces invalid rollouts during subsequent reinforcement learning, since topology diagnostics are meaningful only when the trace contains recoverable reasoning steps.

Stage II: GRPO with TopoPRM Rewards. Starting from the warm-start checkpoint, we optimise the policy with GRPO. For each completion, $\mathcal{E}_\phi$ constructs $\mathcal{H}_y$ and $\mathcal{G}_y$; Eq. 3 then scores the completion. Standard group normalization acts on the complete Choquet reward, so topology can change the ordering of rollouts rather than only break ties inside an outcome stratum. Reward-hacking tests separately measure whether structurally plausible but answer-incorrect traces are over-ranked.

Stage III: Topology-Guided Distillation. The optimized model after Stage II serves as the teacher π_T , while a smaller pretrained checkpoint initializes the student π_S . Topology-Guided Distillation conditions target construction on topology diagnostics from student-generated traces. For each prompt x , the student first rolls out $y_{\text{init}} \sim \pi_S(\cdot | x)$; $\mathcal{E}_\phi$ produces $\mathcal{H}_{y_{\text{init}}}$ and $\mathcal{G}_{y_{\text{init}}}$; a dispatcher ψ then issues a revision instruction targeting a backward edge, cyclic component, orphan conclusion, or continuity break. The teacher generates a revised target under this instruction. Thus topology determines where supervision is requested and which revisions are retained, rather than serving only as a post-hoc score.

Distillation objective. Let $A(y_{\text{init}}, y_{\text{rev}})$ indicate that the revised trace has a correct boxed answer and valid format, satisfies $q_{\text{topo}}^{\text{rev}} \geq \tau_g$, does not reduce either q_{dir} or q_{acyc} relative to the initial trace, and fits a fixed token budget B . The retained dataset is $\mathcal{D}_{\text{TGD}} = \{(x, y_{\text{rev}}) : A(y_{\text{init}}, y_{\text{rev}}) = 1\}$. The student minimises standard teacher-forced distillation loss

$$\mathcal{L}_{\text{distill}}(\theta) = -\mathbb{E}_{(x,y) \sim \mathcal{D}_{\text{TGD}}} \sum_{t=1}^{|y|} \log \pi_S(y_t | x, y_{<t}). \quad (4)$$

Unlike static trace distillation, the target distribution is concentrated on revisions elicited by the current student’s structural failures. The explicit direction and acyclicity constraints prevent apparent compression from being obtained by deleting necessary support or introducing circular justification. The full pipeline is summarised in Appendix Algorithm 1.

4 EXPERIMENTS

Draft audit status. Red cells are not paper claims. Each method uses one canonical seed-0 training run. Parenthesized uncertainty will be a paired per-example bootstrap interval over the same evaluation items; it will not be simulated or described as training variance.

4.1 CANONICAL PROTOCOL

Fixed model scale and budget. The primary comparison starts from the same Qwen3.5-9B Stage-I checkpoint and changes only the Stage-II reward. The public frontier is restricted to dense released checkpoints with 7B–9B parameters; sparse models with similar total but much smaller active parameter counts are outside this matched scale. All variants use rank-64 LoRA, 300 GRPO steps, four generations per prompt, learning rate 5×10^{-6} , $\beta=0.04$, and a 4,096-token completion budget. Stage I uses 18,487 structured math solutions for training and 983 source-disjoint solutions for validation, three epochs, learning rate 5×10^{-5} , and global batch 32. Stage II uses 14,893 unique public train-split questions after exact normalized-question exclusion against every primary evaluation set. We remove 76 of 14,969 leakage-free candidates whose exact Qwen3.5 chat prompts exceed 512 tokens, rather than truncating their problem statements. Reference-solution DAGs are withheld because their step indices are not aligned with generated traces; q_{topo} is computed only from each rollout’s $\mathcal{H}_y$ and $\mathcal{G}_y$. The pair encoder is frozen in every topology-aware run; the 9B semantic teacher is used only to create its offline supervision and never appears in the GRPO reward loop.

Benchmarks and metrics. The primary fixed-scale frontier uses GSM8K, MATH-500, the official 675-item English text-only mathematics subset of OlympiadBench, and AIME 2024, spanning elementary through competition mathematics. Pass@1 exact-answer accuracy and mean generated tokens are co-primary: method A Pareto-dominates B only when A is no less accurate and no longer, with one strict inequality. Pass@5 and maj@5 are secondary diagnostics. Every reproduced row uses one seed-0 sampled response per item with temperature 0.6, top- $p=0.95$, top- $k=20$, min- $p=0$, and repetition penalty 1.0 under an identical instruction, verifier, and response cap. This shared protocol follows the Qwen3 and DeepSeek-R1 reasoning recommendations and deliberately does not mix model-specific best-case decoders; Qwen3.5’s released template remains in thinking mode. Tokenizer and message roles are additionally identical across matched Qwen3.5 rows. OpenReasoning-Nemotron-7B and AceMath-RL-Nemotron-7B receive the same mathematics instruction folded into one user turn, following their released math prompts; Qwen3-8B, DeepSeek-R1-0528-Qwen3-8B, and Klear-Reasoner-8B retain their released templates. In addition to the matched reward controls, we reproduce these checkpoints and the unadapted Qwen3.5-9B checkpoint. Quoted model-card values are context only and are never averaged with reproduced results.

Statistics and provenance. Every reported row maps to a checkpoint, evaluation manifest, and per-example output file. Accuracy differences and length differences use paired bootstrap resampling over shared items. Because training is run once per method, these intervals measure evaluation-set uncertainty only. Several public-frontier checkpoints were released with substantially longer reasoning budgets and repeated decoding. We therefore re-evaluate their released weights under the same single-response 4,096-token protocol rather than transferring quoted scores into Table 1. The human edge audit reports annotator agreement, edge precision/recall, direction accuracy, and abstention coverage separately from teacher-held-out classifier validation.

4.2 ACCURACY–LENGTH FRONTIER (RQ1)

Table 1 tests the central empirical claim: at a fixed 9B adaptation scale and response budget, does the raw-directed TopoPRM reward move the accuracy–length frontier beyond Stage-I SFT, outcome-only GRPO, explicit outcome–format–length shaping, and matched additive or multiplicative multi-source rewards? A SOTA claim will be made only if TopoPRM is non-dominated against all matched

Table 1: Fixed-budget accuracy–length comparison at dense 7B–9B scale. Each cell reports pass@1 accuracy (%) and mean generated tokens from one seed-0 sampled response per item under the same decoder, 4,096-token budget, and task instruction. Released weights and chat templates are retained; each adapted method uses one canonical seed-0 training run. **Pending intervals will come from per-example bootstrap resampling, not training variance.**

Method	GSM8K		MATH-500		OlympiadBench		AIME 2024		Macro avg.	
	Acc.	Tok.	Acc.	Tok.	Acc.	Tok.	Acc.	Tok.	Acc.	Tok.
<i>Public checkpoints, reproduced with the same evaluator</i>										
Qwen3-8B (Qwen Team, 2025)	—	—	—	—	—	—	—	—	—	—
DeepSeek-R1-0528-Qwen3-8B (DeepSeek-AI, 2025)	—	—	—	—	—	—	—	—	—	—
Klear-Reasoner-8B (Su et al., 2025)	—	—	—	—	—	—	—	—	—	—
OpenReasoning-Nemotron-7B (NVIDIA, 2025)	—	—	—	—	—	—	—	—	—	—
AceMath-RL-Nemotron-7B (NVIDIA, 2026)	—	—	—	—	—	—	—	—	—	—
Qwen3.5-9B (Qwen Team, 2026)	—	—	—	—	—	—	—	—	—	—
<i>Matched Qwen3.5-9B adaptation from the same Stage-I checkpoint</i>										
Stage-I SFT	—	—	—	—	—	—	—	—	—	—
Outcome-only GRPO	—	—	—	—	—	—	—	—	—	—
Outcome + format + length GRPO	—	—	—	—	—	—	—	—	—	—
Matched additive reward	—	—	—	—	—	—	—	—	—	—
Matched multiplicative reward	—	—	—	—	—	—	—	—	—	—
TopoPRM (Choquet)	—	—	—	—	—	—	—	—	—	—

Table 2: Mechanism ablations. ΔR_{rev} and ΔR_{cyc} measure the reward drop after direction reversal and cycle injection on matched traces. Hack% is the fraction of answer-incorrect perturbations ranked above their correct source. **All cells await the canonical runs.**

Variant	Acc.	Tok.	ΔR_{rev}	ΔR_{cyc}	Hack%↓
Full TopoPRM	—	—	—	—	—
w/o raw direction	—	—	—	—	—
w/o raw acyclicity	—	—	—	—	—
Projected graph only	—	—	—	—	—
Rules only (no pair encoder)	—	—	—	—	—
w/o continuity	—	—	—	—	—

Table 3: Matched aggregation comparison using the same five normalized sources and training budget. Pareto wins count benchmarks on which a method is not dominated in accuracy and length. **All cells await the canonical runs.**

Aggregation	Acc.	Tok.	Pareto wins	Reward std	Hack%↓
Equal-weight sum	—	—	—	—	—
Matched additive	—	—	—	—	—
Matched multiplicative	—	—	—	—	—
Topology-conditioned Choquet	—	—	—	—	—

rows and improves the reproduced accuracy–length frontier over the listed 7B–9B public checkpoints. Until then, the table remains a preregistered comparison rather than a result narrative.

4.3 WHERE THE GAIN COMES FROM (RQ2)

Raw direction and acyclicity. Table 2 separates our central structural hypothesis from generic reward shaping. We independently remove the backward-edge term, cycle-edge term, both raw-

Table 4: Independent edge audit on direction-balanced traces. All extractor rows are scored against the same adjudicated human edges; Dir. Acc. is conditional on a predicted and annotated edge, and coverage reports the fraction of candidate pairs not abstained on. **Cells await duplicate annotation and adjudication.**

Extractor	Prec.	Rec.	F1	Dir. Acc.	Coverage
Rules only	—	—	—	—	—
Frozen pair encoder	—	—	—	—	—
Hybrid raw-graph encoder	—	—	—	—	—

Table 5: Matched 9B-to-4B distillation protocol. All student methods use the same teacher, candidate prompts, accepted-target count, token budget, and optimizer budget. Accuracy is pass@1; Dir. and Acyc. are paired with the independent edge audit. **Cells remain pending until the single canonical run; intervals will be paired per-example bootstrap intervals, not training variance.**

Method	GSM8K	MATH-500	Avg. Tok	Dir.	Acyc.	Accept%
9B teacher	—	—	—	—	—	—
4B initialization	—	—	—	—	—	—
Static teacher traces	—	—	—	—	—	—
Generic revision	—	—	—	—	—	—
Length-only revision	—	—	—	—	—	—
Topology-Guided Distillation	—	—	—	—	—	—

graph diagnostics, the pair encoder, and continuity. In addition to downstream accuracy and length, controlled graph pairs hold edge count and backward-edge mass fixed while changing whether a backward edge closes a cycle; trace-level reversal holds step text fixed while changing premise/conclusion order. A valid structural signal must distinguish these perturbations before any policy-level gain is attributed to topology.

Non-additive aggregation. Table 3 holds the five sources and compute fixed while changing only their aggregation. The Choquet hypothesis is supported only if topology interactions improve the Pareto frontier or reduce reward hacking beyond a tuned weighted sum and multiplicative control. Reward variance is diagnostic, not a standalone success metric.

4.4 INDEPENDENT STRUCTURAL AUDIT AND COST (RQ3)

The edge audit contains direction-balanced original, reversed, and interleaved traces, with independent duplicate annotation and adjudication. We report pair-level edge precision/recall, conditional direction accuracy, abstention coverage, and inter-annotator agreement. Policy outputs are then audited with two tests: (i) whether high extractor scores predict human-supported edges and (ii) whether answer-incorrect but structurally plausible traces are over-ranked. This prevents the training extractor from serving as its own sole evaluator.

The cost report separates offline semantic labeling and pair-encoder adaptation from online reward computation. For Stage I and Stage II we report wall-clock time, peak per-GPU memory, graph pairs per second, and the fraction of GRPO time spent in graph construction. Source data and large offline artifacts reside on shared object storage. Canonical Stage II checkpoints are saved atomically on the training filesystem, validated as readable safetensors, and only then archived to object storage; logs record the exact $4 \times L20$ allocation.

4.5 TOPOLOGY-GUIDED DISTILLATION (RQ4)

Table 5 asks whether Topology-Guided Distillation moves the 4B accuracy–length frontier beyond static teacher-trace distillation, generic teacher revision, and explicit length-only revision. All methods share the same Stage-II 9B teacher, student initialization, candidate prompts, accepted-target count, response budget, and optimizer budget. Direction and acyclicity retention must survive the independent edge audit rather than only the training extractor. Canonical values will replace the red pending cells after Stage II and the single 4B distillation run complete.

5 CONCLUSION

We presented TopoPRM, a topology-aware process reward model that recovers a raw directed support graph before acyclic projection. The discarded backward and cyclic structure supplies direction and acyclicity supervision; a topology-conditioned Choquet reward couples these diagnostics with outcome, continuity, format, and length; and Topology-Guided Distillation uses the same violations to elicit and filter compact teacher revisions. The policy retains free-form reasoning and requires no graph-format decoding.

LIMITATIONS

TopoPRM provides a structural process signal rather than a proof-level verifier. Consequently, locally plausible yet semantically wrong steps can still receive positive structural credit, especially on competition problems where a single invalid deduction decides the outcome. The raw-graph diagnostics also depend on edge-recovery quality. We therefore report edge precision, direction accuracy, inter-annotator agreement, and abstention coverage rather than treating the encoder as an oracle. Future work can expand the verified-edge set and add uncertainty-aware abstention, couple structural credit with a semantic process verifier, and extend TopoPRM to code, scientific, and multimodal reasoning where executable or simulator-based oracles are available.

REPRODUCIBILITY STATEMENT

This draft is being audited for implementation-to-equation consistency and row-level result provenance. The submission version will point to the exact method definitions, configurations, raw-result manifests, and anonymized code bundle used to reproduce every retained result.

AI USE STATEMENT

Generative AI tools assisted with prior-review analysis, literature search and synthesis, hypothesis and experimental-control refinement, method implementation and code review, dataset cleaning, reproducibility auditing, scientific-figure editing, and manuscript drafting and revision. Before submission, the authors will manually inspect the cited literature, review code and generated artifacts, and verify every reported result against raw outputs. The authors take responsibility for the final claims, code, artifacts, and results; this disclosure will be updated if additional uses occur before submission.

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A IMPLEMENTATION DETAILS

A.1 HYPERPARAMETERS

Table 6 lists the key hyperparameters for each training stage.

Table 6: Hyperparameter summary for the three training stages.

Parameter	Value	Description
<i>Stage I: Supervised Fine-Tuning (SFT)</i>		
Base model	Qwen3.5-9B	–
Adapter	LoRA $r=64$, $\alpha=128$	all-linear modules
Learning rate	5×10^{-5}	cosine, 3% warmup
Epochs	3	–
Batch size	1/device, accum 8	global batch 32 on 4 GPUs
Max seq length	4096	–
Precision	bf16	–
<i>Stage II: GRPO with TopoPRM Rewards</i>		
Base model	SFT checkpoint	–
Adapter	LoRA $r=64$, $\alpha=128$	all-linear modules
Learning rate	5×10^{-6}	cosine, 5% warmup
Gradient steps	300	–
Batch size	1/device, accum 4	global prompt batch 16 on 4 GPUs
Num generations	4 per prompt	for GRPO grouping
Temperature	0.9	rollout sampling
β (KL penalty)	0.04	–
Max completion length	4096	prompt capped at 512
Precision	bf16	–
<i>Stage III: Topology-Guided Distillation</i>		
Teacher model	Stage-II 9B	frozen during distillation
Student model	Qwen3.5-4B	canonical cross-scale run
Adapter	LoRA $r=64$, $\alpha=128$	all-linear modules
Learning rate	2×10^{-5}	cosine, 3% warmup
Epochs	1	–
Batch size	2/device, accum 4	global batch 32 on 4 GPUs
Student temperature	0.7	student-sampled trace
Teacher temperature	0.0	greedy revision target
Target budget	1024 tokens	fixed across methods
Target selection	answer, format, direction, acyclicity	threshold calibrated on validation
Max seq length	4096	–
Precision	bf16	–

A.2 REWARD CONFIGURATION

The Choquet reward in Eq. 3 uses singleton coefficients $(m_o, m_f, m_t, m_c, m_l) = (0.40, 0.05, 0.12, 0.08, 0.03)$ and interactions $(m_{ot}, m_{tc}, m_{tl}) = (0.12, 0.13, 0.07)$. These non-negative coefficients sum to one. The additive and multiplicative alternatives use the same five normalized sources and are tuned under the same budget; the main method uses the complete Choquet reward directly in standard group normalization. In canonical Stage II runs, $\lambda_k=0$: no reference-solution graph is supplied because its nodes are not aligned with a generated trace. Reference-edge alignment remains an optional definition only for settings with explicit node correspondence.

A.3 HARDWARE AND INFRASTRUCTURE

Canonical ICLR runs use $4 \times$ NVIDIA L20 GPUs. We report measured wall-clock time and peak memory from this configuration rather than extrapolating from the earlier A100 runs.

A.4 EVALUATION PROTOCOL

All public benchmark evaluations use the Transformers generation backend. Pass@1 is one sampled response per item from a single seed-0 evaluation pass; this is neither a multi-seed training result nor an average over repeated decodes. The shared decoder uses temperature 0.6, top- $p=0.95$, top- $k=20$, min- $p=0$, and repetition penalty 1.0. This matches Qwen3 and DeepSeek-R1’s reasoning-time recommendations and keeps the public and adapted rows comparable; it differs from Qwen3.5’s general-task recommendation only in using temperature 0.6 instead of 1.0. The four primary mathematics benchmarks use the same 4,096-token completion cap and the same instruction to provide a step-by-step solution with the final answer in `\boxed{}`. Released chat templates are retained. For OpenReasoning-Nemotron-7B and AceMath-RL-Nemotron-7B, the instruction is folded into one user turn to match their released mathematics prompts; no model receives an uncounted response prefix. Answer extraction takes the last boxed expression, and `math_verify` checks symbolic equivalence. Pass@5 and maj@5, when run as secondary diagnostics, draw five samples from the same decoder and reuse this equivalence relation rather than raw string equality. The manifest stores the message-role protocol, prefix flag, decoder parameters, evaluation seed, batch size, model and adapter paths, dataset fingerprint, and full per-example responses; cached results are reused only when these protocol fields match.

B CHOQUET CAPACITY AND BASELINES

For non-negative Möbius coefficients summing to one, Eq. 3 is monotone in every source, maps 0 to 0 and 1 to 1, and remains in $[0, 1]$. The topology interaction mass $m_{ot} + m_{tc} + m_{tl}$ is separately ablated to distinguish the proposed non-additive mechanism from simply increasing topology’s singleton weight. We also test traces that are structurally plausible but answer-incorrect to detect reward hacking.

C DAG EXTRACTION DETAILS

Extraction pipeline. Given a reasoning trace in `<think>`, we apply five frozen stages:

1. Step segmentation: split by numbered-step markers, mathematical discourse markers (`:`, `;`, `.`), and line boundaries.
2. Sub-question partitioning: markers such as “Sub-question k ” define connected components; cross-sub-question edges are disallowed.
3. Information extraction: for each step, extract expressions, claims, and variables with rule-based patterns.
4. Raw support encoding: combine high-precision symbolic candidates with a lightweight pair classifier that selects $i \rightarrow j$, $j \rightarrow i$, or abstention without imposing text order.
5. Projection: retain the raw graph for diagnostics, remove backward edges to obtain the training DAG, then add adaptive weak sequential edges only when support is absent.

Information-density filters. To avoid low-information nodes and noisy links: (a) claims are extracted at sentence level, with heading fragments and unfinished prefixes removed; (b) trivial tokens (single symbols, pure numerals, non-relational fragments) are excluded from dependency matching.

Dependency rules. For each unordered pair of steps in the same sub-question, rules propose candidates from informative expression reuse, canonicalized claim overlap, variable roles, and explicit step references. The pair classifier assigns direction or abstains. Only after raw-graph diagnostics are computed do we insert a fallback edge for an unsupported derivation or conclusion in the projected DAG.

Pair-encoder adaptation. The lightweight encoder is Qwen3.5-0.8B with a rank-8 LoRA three-way classification head. Offline supervision contains 540 usable source traces labeled by the frozen 9B semantic teacher. Each source trace is presented in original, reversed, and interleaved step order, while the train-validation split is performed before augmentation at the source-trace level. The

encoder is frozen during policy optimization; its confidence and direction-margin abstention thresholds are calibrated on the held-out teacher split and audited separately against human edge labels. A retained encoder edge keeps its calibrated confidence as w_{ij} ; auditable rule edges use $w_{ij} = 1$.

Adaptive sequential weak edges. A weak sequential edge (weight 0.3) is added only when the target step lacks strong incoming support and the source step does not already emit strong outgoing dependencies. This prevents chain-dominant artifacts in long traces.

Node typing. Each node is classified into DEFINITION, DERIVATION, COMPUTATION, CONCLUSION, AUXILIARY, SUBSTITUTION, or CASE_ANALYSIS by keyword rules.

Relationship between DAG quality and reasoning compression. We observe empirically that traces with higher DAG scores (acyclicity, low orphan ratio, strong directional consistency) tend to admit shorter compressed forms under Topology-Guided Distillation. Intuitively, a well-structured DAG has fewer redundant branches and shorter critical paths, so the transitive reduction yields a more compact chain. The topology source q_{topo} enters the Choquet reward in Eq. 3, while the projected DAG supplies the compression target.

D REWARD FUNCTION DETAILS

Outcome reward. Extracts the final mathematical answer from the last $\boxed{}$ expression, falling back to an explicit final numeric expression when needed. Symbolic equivalence to the reference answer is checked with `math_verify`; a correct answer receives 1 and an incorrect or unparseable answer receives 0.

Format reward. Scores recognizable mathematical reasoning and conclusion structure: a `<think>` block followed by a boxed answer receives 1.0; multi-step reasoning with a boxed answer receives 0.8; a boxed answer alone receives 0.5; a `<think>` block without a boxed conclusion receives 0.3; and unboxed step-by-step reasoning receives 0.1. Empty or unstructured output receives 0.

Length reward. Linear decay from 1.0 (at $\leq 2,000$ characters) to 0.0 (at $\geq 8,000$ characters). Evaluation tables report tokenizer-token counts independently of this training-time shaping signal.

Topology reward (r_{topo}). The topology source used in GRPO is computed jointly from the raw support graph and projected DAG (Eq. 5). It aggregates graph existence, coverage-adjusted acyclicity, absence of orphan conclusions, coverage-adjusted direction consistency, and optional reference-edge alignment.

Continuity reward. Uses regex-based expression matching to verify that each step references expressions or claims from prior steps or the problem statement. The continuity score q_{cont} (Eq. 6) measures the fraction of supported steps.

E SUPPLEMENTARY MATHEMATICAL DEFINITIONS

Topology score (full form). Let c_H be the fraction of nodes incident to a dependency edge in the raw graph, and let d_H and u_H be the confidence-weighted forward-edge and cycle-edge mass defined in Eqs. (1)–(2). Coverage-adjusted diagnostics treat missing edge evidence as uncertainty:

$$\begin{aligned} q_{\text{topo}}(\mathcal{H}, \mathcal{G}) = & \sigma(\lambda_b \mathbb{I}[|\mathcal{V}| > 0] \\ & + \lambda_a [c_H(1 - u_H) + (1 - c_H)/2] \\ & + \lambda_o \mathbb{I}[\rho_{\text{orphan}}(\mathcal{G}) = 0] \\ & + \lambda_d [c_H d_H + (1 - c_H)/2] \\ & + \lambda_k \kappa(\mathcal{G}, \mathcal{G}^*)), \end{aligned} \tag{5}$$

where $\sigma(\cdot)$ normalizes by the active weights, ρ_{orphan} is orphan-conclusion ratio on the projected DAG, and κ is optional reference-edge alignment. The terms in Eq. (5) are aggregated with weights

Algorithm 1 Topology-Guided Distillation Training Pipeline

Require: Base policy π_0 ; data $\mathcal{D}_{\text{sft}}, \mathcal{D}_{\text{rl}}$; Choquet coefficients $\{m_i, m_{ij}\}$

Ensure: Teacher π_T , compact student π_S

```

1:  $\pi_\theta \leftarrow \text{SFT}(\pi_0, \mathcal{D}_{\text{sft}})$  ▷ Stage I
2: for each GRPO step on  $\mathcal{D}_{\text{rl}}$  do ▷ Stage II
3:   Sample  $\{y^{(i)}\}_{i=1}^G \sim \pi_\theta(\cdot | x)$ ; build  $\mathcal{H}^{(i)} = \mathcal{E}_\phi(y^{(i)})$  and  $\mathcal{G}^{(i)} = \Pi(\mathcal{H}^{(i)})$ 
4:   Compute  $r_{\text{out}}^{(i)}, r_{\text{fmt}}^{(i)}, r_{\text{len}}^{(i)}, q_{\text{topo}}^{(i)}, q_{\text{cont}}^{(i)}$ 
5:   Compute  $r_{\text{total}}^{(i)}$  by Eq. 3; normalize within the GRPO group
6:   Update  $\pi'_\theta$  with the GRPO objective
7: end for
8:  $\pi_T \leftarrow \pi'_\theta$ ; init.  $\pi_S$  from a smaller pretrained checkpoint
9: for each prompt  $x \in \mathcal{D}_{\text{rl}}$  do ▷ Stage III
10:   $y_{\text{init}} \sim \pi_S(\cdot | x)$ ; extract defect record  $F(y_{\text{init}})$  from  $\mathcal{H}$  and  $\mathcal{G}$ 
11:   $P_r \leftarrow \psi(F(y_{\text{init}}))$ ; sample  $y_{\text{rev}} \sim \pi_T(\cdot | x, y_{\text{init}}, P_r)$ 
12:  Retain  $(x, y_{\text{rev}})$  only if answer, format, direction, acyclicity, and budget constraints pass
13:  Update  $\pi_S$  with  $\mathcal{L}_{\text{distill}}$  (Eq. 4)
14: end for
15: return  $(\pi_T, \pi_S)$ 

```

$(\lambda_b, \lambda_a, \lambda_o, \lambda_d, \lambda_k) = (0.20, 0.15, 0.15, 0.15, 0.25)$. The normalizer σ divides by the sum of active weights; when no reference graph exists, λ_k is removed before normalization. This guarantees $q_{\text{topo}} \in [0, 1]$ while preventing an empty raw graph from receiving perfect direction and acyclicity credit.

Continuity score (full form). The local continuity quality is

$$q_{\text{cont}}(y) = \begin{cases} 1.0, & \eta(y) = 1, \\ 0.8\eta(y), & \eta(y) < 1, \end{cases} \quad \eta(y) = \frac{1}{T} \sum_{i=1}^T \mathbb{1}[\text{supported}(s_i, s_{<i}, q)]. \quad (6)$$

Remark. Eq. (6) is the engineering-tuned continuity term used in the implementation. It is a separate source in the Choquet reward rather than being folded into topology before aggregation.

Topology-conditioned revision dispatch. The revision instruction is generated by

$$P_r = \psi(r_{\text{out}}, q_{\text{dir}}, q_{\text{acyc}}, \rho_{\text{orphan}}, q_{\text{cont}}, \ell), \quad (7)$$

where ℓ identifies the offending edge, cyclic component, or unsupported step when one can be localized.

F TOPOLOGY-GUIDED DISTILLATION DETAILS

Topology-Guided Distillation is an optional compression stage whose goal is to retain the teacher’s structural reasoning profile while reducing generation length in compact students.

Why static trace distillation is insufficient. Conventional trace distillation (Hinton et al., 2015; Gu et al., 2024) presents every student with the same teacher solutions. It therefore cannot target a student’s particular missing dependency, and a length-only target can become shorter by deleting support that is necessary for the conclusion. Topology-Guided Distillation instead constructs supervision after observing the compact student’s own trace.

Defect localization and revision. For each initial student attempt, we retain the raw edge list and projected DAG. The defect record contains the endpoints of backward edges, nodes in a non-trivial strongly connected component, orphan conclusions in the projected DAG, and unsupported local transitions. The dispatcher selects the highest-priority localized defect and asks the frozen Stage-II teacher to rewrite the solution within a fixed response budget while preserving a boxed final answer.

Target acceptance. A revised trace is retained only if it has a correct boxed answer and valid format, reaches the calibrated topology threshold, does not reduce coverage-adjusted direction consistency or acyclicity relative to the initial trace, and stays within the shared token budget. We log every rejection reason and report acceptance coverage. This makes direction and acyclicity part of the distillation data contract, not merely an evaluation diagnostic.

Student training and controls. The 4B student is trained on accepted revisions with Eq. 4. Matched controls use the same teacher, prompts, number of accepted targets, and optimization budget but replace topology-conditioned dispatch with either a generic revision instruction or an explicit length-only instruction. External answer accuracy and generated tokens are primary; extractor-based structural retention is reported as internal-consistency evidence and paired with the independent human edge audit.